The Largest Sidon Subset of the Perfect Squares $\{1^2, 2^2, \ldots, N^2\}$

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Abstract

A *Sidon set* (also called a B_2 set) is a set of integers in which all pairwise sums are distinct. Let $F(N)$ denote the maximum size of a Sidon subset of $\mathcal{S}_N = \{1^2, 2^2, \ldots, N^2\}$. We prove

$$\Omega\left(\frac{N^{2/3}}{(\log N)^{1/3}}\right) \;\leq\; F(N) \;\leq\; O\left(\frac{N}{(\log N)^{1/4}}\right).$$

In particular $F(N) = \omega(N^{1/2})$. The upper bound uses the Landau–Ramanujan theorem on the density of sums of two squares. The lower bound is a probabilistic alteration construction. The question of whether $F(N) = N^{1-o(1)}$ is not settled by these bounds.

Contents

1 Introduction and Statement of Results

Definition 1.1. A set A of integers is a *Sidon set* (or B_2 *set*) if whenever $a, b, c, d \in A$ satisfy $a + b = c + d$ we have $\{a, b\} = \{c, d\}$ as multisets.

Let $N \geq 1$ and define

$$\mathcal{S}_N \;=\; \{k^2 : 1 \leq k \leq N\}, \quad |\mathcal{S}_N| = N.$$

Set

$$F(N) \;:=\; \max\bigl\{|A| : A \subseteq \mathcal{S}_N,\ A \text{ is Sidon}\bigr\}.$$

Theorem 1.2 (Main Theorem)**.**

$$\Omega\!\left(\frac{N^{2/3}}{(\log N)^{1/3}}\right) \;\leq\; F(N) \;\leq\; O\!\left(\frac{N}{(\log N)^{1/4}}\right).$$

Corollary 1.3. $F(N) = \omega(N^{1/2})$.

Remark 1.4. Note that $N/(\log N)^{1/4} = N^{1-o(1)}$ (since the exponent $1 - \frac{\log\log N}{4\log N} \to 1$), so the upper bound $O(N/(\log N)^{1/4})$ is entirely consistent with $F(N) = N^{1-o(1)}$. The question of whether $F(N) = N^{1-o(1)}$ is *not settled* by Theorem 1.2.

1.1 Historical context

The classical Sidon problem asks for the maximum size of a Sidon subset of $\{1, \ldots, M\}$; the answer is $(1 + o(1))M^{1/2}$ (Erdős–Turán upper bound [1], Singer difference set lower bound [4]). Our host set $\mathcal{S}_N$ has N elements and lies in $\{1, \ldots, N^2\}$, and the Sidon condition on sums $a_i^2 + a_j^2$ interacts with the analytic structure of the two-squares representation function r_2.

2 Preliminaries

Definition 2.1. For $n \geq 0$ define

$$r_2(n) = \#\{(a,b) \in \mathbb{Z}^2 : a^2+b^2 = n\}, \quad r_2^+(n; N) = \#\{(a,b) \in \mathbb{N}^2 : a^2+b^2 = n,\ 1 \leq a \leq b \leq N\}.$$

Lemma 2.2 (Second moment of r_2; [2, Ch. 18])**.** *As $M \to \infty$,*

$$\sum_{n=0}^{M} r_2(n)^2 \;\sim\; 4\pi M \log M.$$

Proof. The sum equals the number of 4-tuples $(a,b,c,d) \in \mathbb{Z}^4$ with $a^2 + b^2 = c^2 + d^2 \leq M$. Via the factorisation $r_2(n) = 4(d_1(n) - d_3(n))$ (where d_i counts divisors $\equiv i \pmod 4$) and standard Dirichlet series methods, one obtains the stated asymptotic; see [2, Theorem 339]. □

Lemma 2.3 (Landau–Ramanujan; [3])**.** *As $M \to \infty$,*

$$\#\{n \leq M : r_2(n) > 0\} \;\sim\; K \cdot \frac{M}{\sqrt{\log M}},$$

where $K = \frac{1}{\sqrt{2}} \prod_{p\equiv 3(4)} (1 - p^{-2})^{-1/2} \approx 0.7642\ldots$ is the Landau–Ramanujan constant.

An integer n is a sum of two squares if and only if every prime $p \equiv 3 \pmod 4$ divides n to an even power; see [2, Thm. 366]. Lemma 2.3 follows from this characterisation via a standard Euler product computation and a Tauberian theorem.

3 Upper Bound

Theorem 3.1. $F(N) \leq O\big(N/(\log N)^{1/4}\big)$.

Proof. Let $A = \{a_1^2, \ldots, a_k^2\} \subseteq \mathcal{S}_N$ be a Sidon set with $1 \leq a_1 < \cdots < a_k \leq N$ and $k = |A|$.
Step 1. Since A is Sidon, the $\binom{k+1}{2}$ values

$$\Sigma(A) = \big\{a_i^2 + a_j^2 : 1 \leq i \leq j \leq k\big\}$$

are pairwise distinct. Each element of $\Sigma(A)$ is a sum of two positive squares bounded by N, so

$$\Sigma(A) \;\subseteq\; R_N := \{n \leq 2N^2 : r_2(n) > 0\}.$$

Step 2. By Lemma 2.3 with $M = 2N^2$,

$$|R_N| \;\leq\; K \cdot \frac{2N^2}{\sqrt{\log(2N^2)}} \cdot (1 + o(1)) \;=\; \frac{2K\,N^2}{\sqrt{2\log N}} \cdot (1 + o(1)) \;=\; \frac{K\sqrt{2}\,N^2}{\sqrt{\log N}} \cdot (1 + o(1)).$$

Step 3. Since all $\binom{k+1}{2}$ elements of $\Sigma(A)$ lie in R_N,

$$\frac{k(k+1)}{2} \;\leq\; |R_N| \;\leq\; C\,\frac{N^2}{(\log N)^{1/2}}$$

for an absolute constant $C > 0$. Hence $k^2 \leq 2C\,N^2/(\log N)^{1/2}$, giving $k \leq \sqrt{2C}\,N/(\log N)^{1/4}$. □

4 Lower Bound

Theorem 4.1. $F(N) \geq \Omega\big(N^{2/3}/(\log N)^{1/3}\big)$.

We use the probabilistic method with alteration.

Definition 4.2. A *collision* is an ordered quadruple $(a, b, c, d) \in \{1, \ldots, N\}^4$ with $a < b$, $c < d$, $\{a, b\} \neq \{c, d\}$, and $a^2 + b^2 = c^2 + d^2$. Let $\mathcal{C}$ be the set of all collisions.

Lemma 4.3. $|\mathcal{C}| \leq 8\pi N^2 \log N \cdot (1 + o(1))$.

Proof. For each $n \in \{2, \ldots, 2N^2\}$, the collisions at level n number $\binom{r_2^+(n;N)}{2} \leq \frac{1}{2} r_2^+(n; N)^2 \leq \frac{1}{2} r_2(n)^2$. Summing and applying Lemma 2.2 with $M = 2N^2$:

$$|\mathcal{C}| \;\leq\; \frac{1}{2}\sum_{n=0}^{2N^2} r_2(n)^2 \;\sim\; \frac{1}{2} \cdot 4\pi \cdot 2N^2 \cdot \log(2N^2) \;\sim\; 8\pi N^2 \log N. \qquad \square$$

Proof of Theorem 4.1. Let $\lambda > 0$ be a parameter to be chosen. Include each $a \in \{1, \ldots, N\}$ independently in a random set $\mathcal{A}$ with probability λ. Let $X = |\mathcal{A}|$ and let Y be the number of collisions in $\mathcal{A}$.

Expectations:

$$\mathbb{E}[X] = N\lambda, \qquad \mathbb{E}[Y] = |\mathcal{C}|\lambda^4 \le 8\pi N^2 \log N \cdot \lambda^4 \cdot (1 + o(1)).$$

Alteration: For each collision in $\mathcal{C}$ whose four elements all lie in $\mathcal{A}$, mark one of the four for deletion; then delete the union of all marked elements from $\mathcal{A}$, obtaining a surviving index set $\mathcal{A}^*$. Because every collision is hit by this procedure, the square set $\{a^2 : a \in \mathcal{A}^*\}$ is a Sidon subset of $\mathcal{S}_N$. At most one element is deleted per collision, so $|\mathcal{A}^*| \ge X - Y$. Hence

$$\mathbb{E}[|\mathcal{A}^*|] \;\ge\; N\lambda - 8\pi N^2 \log N \cdot \lambda^4 \cdot (1 + o(1)).$$

Optimising λ: Maximise $g(\lambda) = N\lambda - 8\pi N^2 \log N \cdot \lambda^4$. Setting $g'(\lambda) = N - 32\pi N^2 \log N \cdot \lambda^3 = 0$ gives

$$\lambda^* \;=\; \left(\frac{1}{32\pi N \log N}\right)^{1/3} \;=\; \frac{1}{(32\pi)^{1/3}\, N^{1/3} (\log N)^{1/3}}.$$

Since $8\pi N \log N \cdot (\lambda^*)^3 = 8\pi N \log N \cdot \frac{1}{32\pi N \log N} = \frac{1}{4}$, we get

$$g(\lambda^*) = N\lambda^*\big(1 - 8\pi N \log N \cdot (\lambda^*)^3\big) = N\lambda^*\big(1 - \tfrac{1}{4}\big) = \frac{3}{4}\, N\lambda^* = \frac{3}{4(32\pi)^{1/3}} \cdot \frac{N^{2/3}}{(\log N)^{1/3}}.$$

Conclusion: With $\lambda = \lambda^*$,

$$\mathbb{E}[|\mathcal{A}^*|] \;\ge\; \frac{3}{4(32\pi)^{1/3}} \cdot \frac{N^{2/3}}{(\log N)^{1/3}} \cdot (1 + o(1)).$$

Since the expectation is an average, there exists a realisation with

$$|\mathcal{A}^*| \;\ge\; \mathbb{E}[|\mathcal{A}^*|] \;\ge\; \frac{3}{4(32\pi)^{1/3}} \cdot \frac{N^{2/3}}{(\log N)^{1/3}} \cdot (1 + o(1)),$$

giving $F(N) \ge \Omega(N^{2/3}/(\log N)^{1/3})$. □

5 Proof of the Main Results

Proof of Theorem 1.2. The lower bound is Theorem 4.1. The upper bound is Theorem 3.1. □

Proof of Corollary 1.3. Theorem 4.1 gives $F(N) \ge \Omega(N^{2/3}/(\log N)^{1/3})$. Since $N^{2/3}/(\log N)^{1/3} \gg N^{1/2}$ (their ratio $N^{1/6}/(\log N)^{1/3} \to \infty$), this is $\omega(N^{1/2})$. □

6 Discussion and Open Problems

6.1 Summary

We have proved:

1. $F(N) = \omega(N^{1/2})$: the maximum Sidon subset of $\mathcal{S}_N$ grows strictly faster than $N^{1/2}$.
2. $F(N) \le O(N/(\log N)^{1/4})$: it grows no faster than N up to a logarithmic factor.

The question of whether $F(N) = N^{1-o(1)}$ is not resolved. The upper bound $O(N/(\log N)^{1/4})$ is itself $N^{1-o(1)}$ (since $(\log N)^{1/4} = N^{o(1)}$, so the exponent of N approaches 1), and thus does not rule out $F(N) = N^{1-o(1)}$. The lower bound $\Omega(N^{2/3}/(\log N)^{1/3})$ is $N^{2/3+o(1)}$, which is consistent with the true answer being either $N^{1-o(1)}$ or strictly smaller. Closing this gap is the central open problem.

6.2 Sharpness

- The Landau–Ramanujan bound in Step 2 of the upper bound proof is sharp: the number of sums-of-two-squares up to M is indeed $\Theta(M/(\log M)^{1/2})$.
- The alteration method is flexible but not tight; a construction via Sidon sets in $\mathbb{F}_p$ or difference sets could potentially give a better explicit lower bound.

6.3 Open problem

- Determine the exponent α such that $F(N) = N^{\alpha+o(1)}$. If $F(N) = N^{\alpha+o(1)}$ for some exponent α, then the bounds are consistent with $\alpha \in [2/3, 1]$.
- Find an explicit (deterministic) construction of a Sidon subset of $\mathcal{S}_N$ of size $\Omega(N^{2/3})$.

References

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